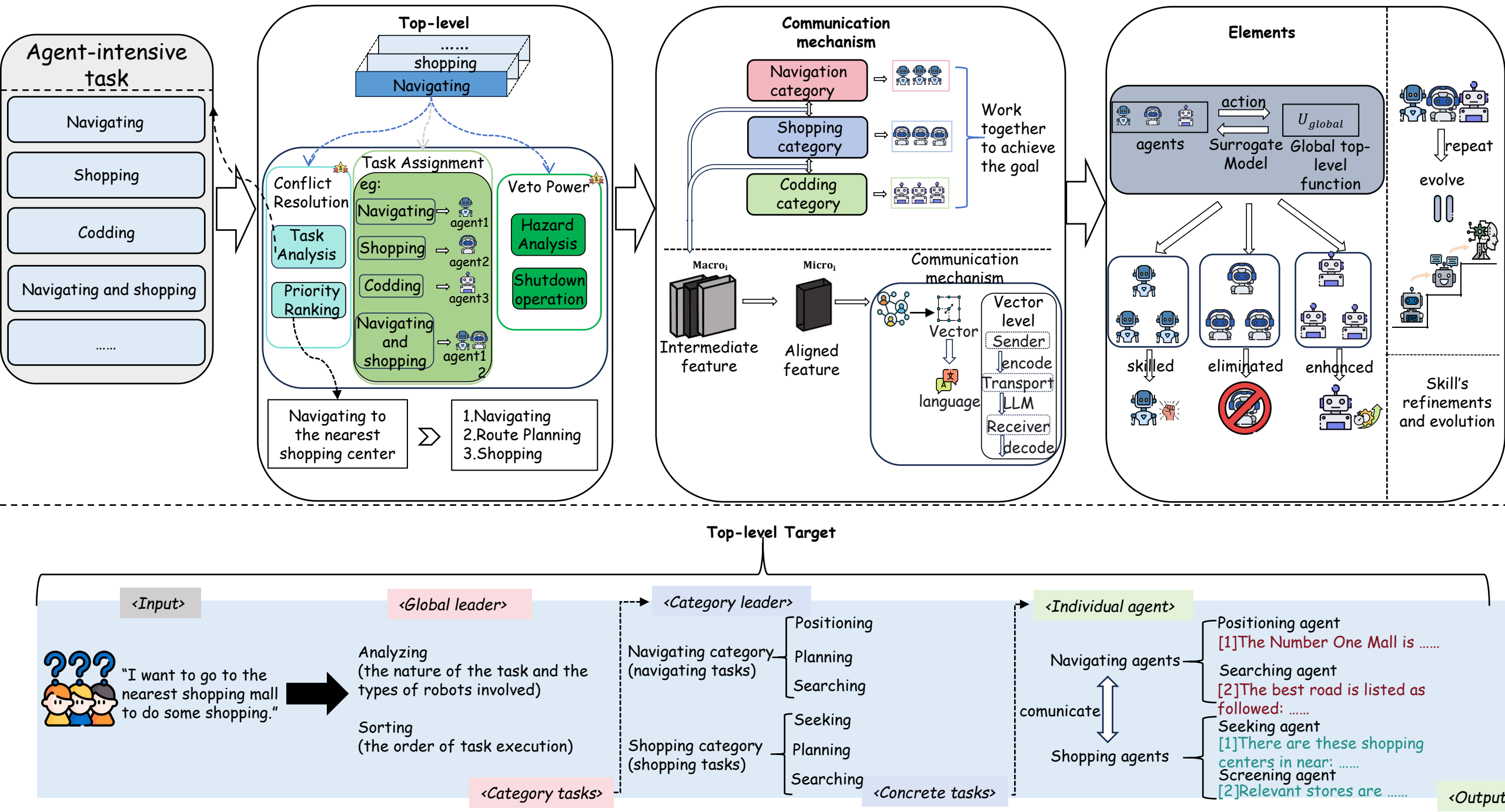
Agent-intensive task
Navigating
Shopping
Codding
Navigating and shopping
.....
Top-level
.....
shopping
Navigating
Conflict Resolution
Task Analysis
Priority Ranking
Task Assignment
eg:
Navigating
agent1
Shopping
agent2
Codding
agent3
Navigating and shopping
agent1 2
Veto Power
Hazard Analysis
Shutdown operation
Navigating to the nearest shopping center
1.Navigating
2.Route Planning
3.Shopping
Communication mechanism
Navigation category
Shopping category
Codding category
Work together to achieve the goal
Macro$_i$
Micro$_i$
Intermediate feature
Aligned feature
Communication mechanism
Vector
language
Vector level
Sender
encode
Transport
LLM
Receiver
decode
Elements
action
agents
Surrogate Model
U_{global}
Global top-level function
skilled
eliminated
enhanced
repeat
evolve
Skill's refinements and evolution
Top-level Target
<Input>
"I want to go to the nearest shopping mall to do some shopping."
<Global leader>
Analyzing
(the nature of the task and the types of robots involved)
Sorting
(the order of task execution)
<Category tasks>
<Category leader>
Navigating category
(navigating tasks)
Positioning
Planning
Searching
Shopping category
(shopping tasks)
Seeking
Planning
Searching
<Concrete tasks>
<Individual agent>
Navigating agents
Positioning agent
[1]The Number One Mall is .....
Searching agent
[2]The best road is listed as followed: .....
comunicate
Shopping agents
Seeking agent
[1]There are these shopping centers in near: .....
Screening agent
[2]Relevant stores are .....
<Output